



ALLIANCE
UNIVERSITY

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ALLIANCE SCHOOL OF ADVANCED COMPUTING

Mini Project: Zomato Data Analysis

6CS 1039

BIG DATA ANALYTICS

Student's Names

Saya Rishi-2022BCSE07AED028

Saya Preetham-2022BCSE07AED029

Gadamsetty Deepthika-2022BCSE07AED110

Araveti Vishnu- 2022BCSE07AED127

Sadineni Aasritha-2022BCSE07AED139

Lagisetty Bhupala Vignesh-2022BCSE07AED283

Instructor's Name:

Dr. Jayabhaduri R

Submission Date:

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**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING WITH
SPECIALIZATION IN DATA ANALYTICS**

S. No	CONTENTS	PAGE NO
1	INTRODUCTION	3
2	PROBLEM DEFINITION AND DATASET SOURCE	3-4
3	BIG DATA FRAMEWORK DETAILS	5
4	IMPLEMENTATION	6-11
5	CONCLUSION	11
6	REFERENCES	12

1. INTRODUCTION

The restaurant industry had a significant impact on how people behave in today's data-driven culture. One of the leading platforms to discover restaurants across the globe, Zomato connects customers with restaurants through ratings, feedback, menus, and additional information. Restaurants which want to improve their menu items need an extensive knowledge of customer habits, prices, and dining preferences. In order to gain understanding of restaurant trends, customer preferences, and operational indicators, this study examines a large dataset from Zomato. We use Python for visualization and exploratory data analysis (EDA) and SQL for structured queries.

2. PROBLEM DEFINITION

Background

Users may find it challenging to make well-informed decisions because of the huge variety of restaurants and cuisine alternatives available on locations such as Zomato. By exposing actionable trends in customer behavior and restaurant performance, data analytics, especially when combined with structured and visual tools, has the potential to allay these worries.

Objectives

1. In order to solve critical business and user-centric issues, this project will investigate the Zomato dataset:
2. Discover which cuisines were most popular in every nation.
3. Find out which cities have the best eateries.
4. Examine how services like table reservations and online delivery affect customer evaluations.
5. Recognize the relationship between ratings and popularity and average cost and pricing range.

Challenges

- Handling inconsistent or missing data, especially with user reviews and ratings
- standardize different currency types across countries
- Interpreting fields having various values, like cuisines
- Make sure that significant categorical features (such location and cuisines) are visualized in a meaningful way
- Accurately joining datasets, particularly with country codes

Data Collection and Preparation

The system processes data from two sources:

- Zomato data (Zomato.csv)
- Country Code data (Country-Code.xlsx)

- **zomato.csv**: This file contains over 9,000 entries from various countries, including:

Restaurant ID and Name

Location: Country Code, City, Address, Locality

Features: Cuisines, Average Cost for Two, Price Range, Currency

User engagement: Votes, Ratings

Services: Has Table Booking, Has Online Delivery, Is Delivering Now

- **Country-Code.xlsx**: A supplementary file mapping numeric country codes to full country names, which helps enhance the dataset for analysis.

The dataset includes both categorical and numerical attributes and offers a great opportunity for performing SQL queries and Python-based EDA.

Libraries Used in the Project:

Data Manipulation & Analysis:

- **pandas** – For loading, cleaning, and manipulating the dataset.
- **numpy** – For numerical operations and working with arrays.

Data Visualization:

- **matplotlib.pyplot** – For plotting charts like bar graphs and pie charts.
- **seaborn** – For enhanced visualizations like heatmaps and boxplots

3. Big Data Framework Details

Overview

In this project, although we are not using a distributed computing framework like Hadoop or Spark, we approach the Zomato dataset using concepts aligned with the big data lifecycle: data ingestion, processing, querying, and visualization. The tools and techniques chosen to provide a scalable foundation suitable for future big data integration.

Framework Components

Component	Description
Data Source	Zomato dataset (CSV) + Country Code (Excel) — structured, semi-clean data
Data Processing	Performed using Python (pandas, NumPy) for cleaning, merging, formatting
Data Management	MySQL used to create schema and query data via SQL
Visualization Layer	Jupyter Notebook with matplotlib, seaborn, and plotly
Storage	Local storage; can be extended to cloud-based solutions like S3, GCS

4. IMPLEMENTATION

4.1 SQL Implementation

The analysis began with creating a SQL database named ZomatoDB and a table zomato_dataset. The schema included fields like RestaurantID, RestaurantName, City, Cuisines, Currency, Price_range, Votes, and Rating.

SQL Queries Used:

- CREATE DATABASE ZOMATODB;
- USE ZOMATODB;
- CREATE TABLE zomato_dataset (
 RestaurantID INT PRIMARY KEY,
 RestaurantName VARCHAR(255),
 CountryCode VARCHAR(100),
 City VARCHAR(100),
 Address VARCHAR(100),
 Locality VARCHAR(255),
 LocalityVerbose VARCHAR(100),
 Cuisines TEXT,
 Currency VARCHAR(50),
 Has_Table_booking VARCHAR(10),
 Has_Online_delivery VARCHAR(10),
 Is_delivering_now VARCHAR(20),
 Switch_to_order_menu VARCHAR(20),
 Price_range INT,
 Votes INT,
 Average_Cost_for_two INT,
 Rating DECIMAL(3,1)
);

Ran basic data exploration queries:

- SELECT DISTINCT CountryCode FROM zomato_dataset;
- SELECT DISTINCT City FROM zomato_dataset;
- SELECT DISTINCT Cuisines FROM zomato_dataset;
- SELECT DISTINCT Currency FROM zomato_dataset;

	CountryCode
►	1
	189
	214
	191
	208
	215
	166
	162
	30
	148
	94
	14
	37
	216
	184

Fig1- Country Codes

	City
►	New Delhi
	Gurgaon
	Noida
	Ghaziabad
	Faridabad
	Pune
	Ludhiana
	Kolkata
	Mumbai
	Bangalore
	Chennai
	Pretoria
	Secunder...
	Hyderabad
	Kochi
	Jaipur
	Ahmedabad
	Chandigarh

Fig2- City

	Cuisines
►	North Indian Chinese Mughlai
	Chinese Thai
	North Indian Mughlai
	Italian
	Chinese Seafood Thai
	North Indian Continental European
	Chinese
	Chinese Thai Seafood
	Chinese Thai Japanese
	South Indian
	Chinese North Indian
	North Indian
	Chinese Seafood
	Chinese Seafood Asian
	North Indian Mughlai Chinese
	Pizza Fast Food
	American Fast Food Salad Health...
	Fast Food Burger

Fig3- Cuisines

	Currency
►	Indian Rupees(Rs.)
	Rand(R)
	Emirati Diram(AED)
	Sri Lankan Rupee(LKR)
	Turkish Lira(TL)
	Pounds(?)
	Qatari Rial(QR)
	Botswana Pula(P)
	Brazilian Real(R\$)
	NewZealand(\$)
	Indonesian Rupiah(IDR)
	Dollar(\$)

Fig4- Currency

4.2 Python EDA and Visualizations

```
# Load CSV data
zomato = pd.read_csv("zomato (1).csv", encoding='latin-1')

# Load country code data
country = pd.read_excel("Country-Code.xlsx")
```

Visualization:

We used matplotlib, seaborn, and plotly to create various visualizations.

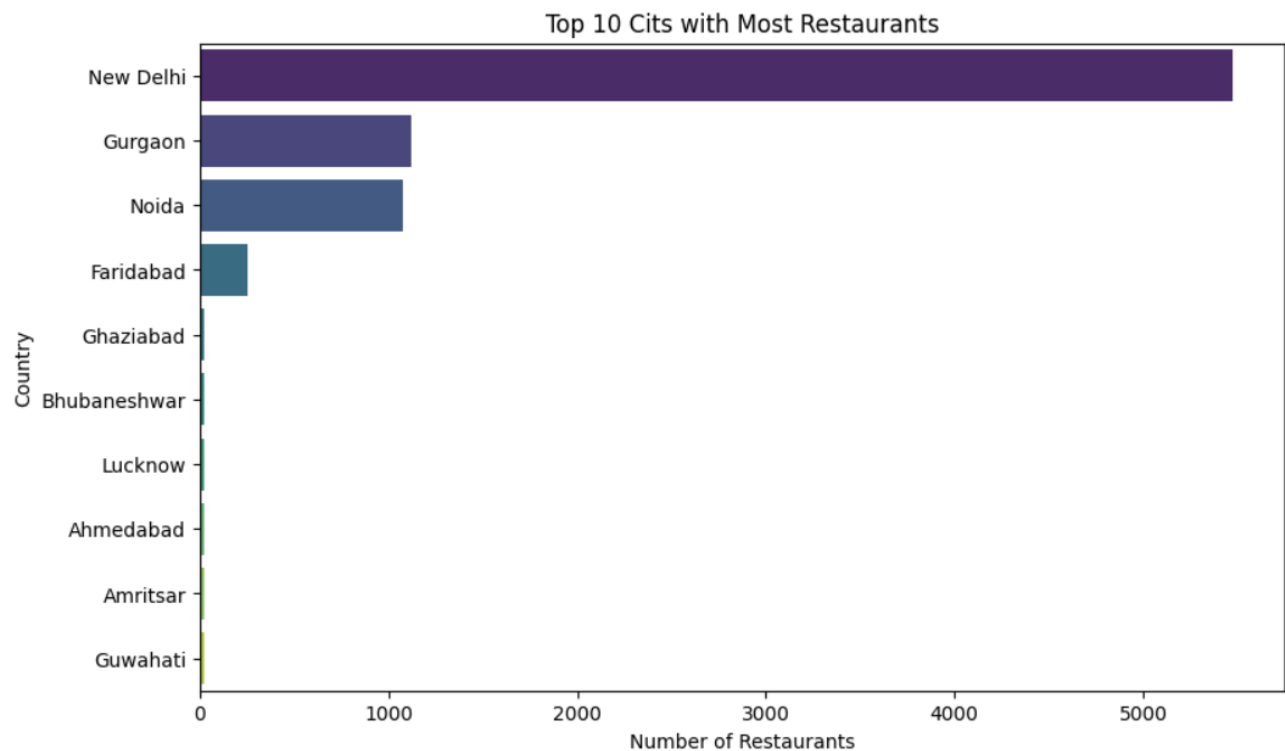


Fig5-Top 10 Countries with Most Restaurants

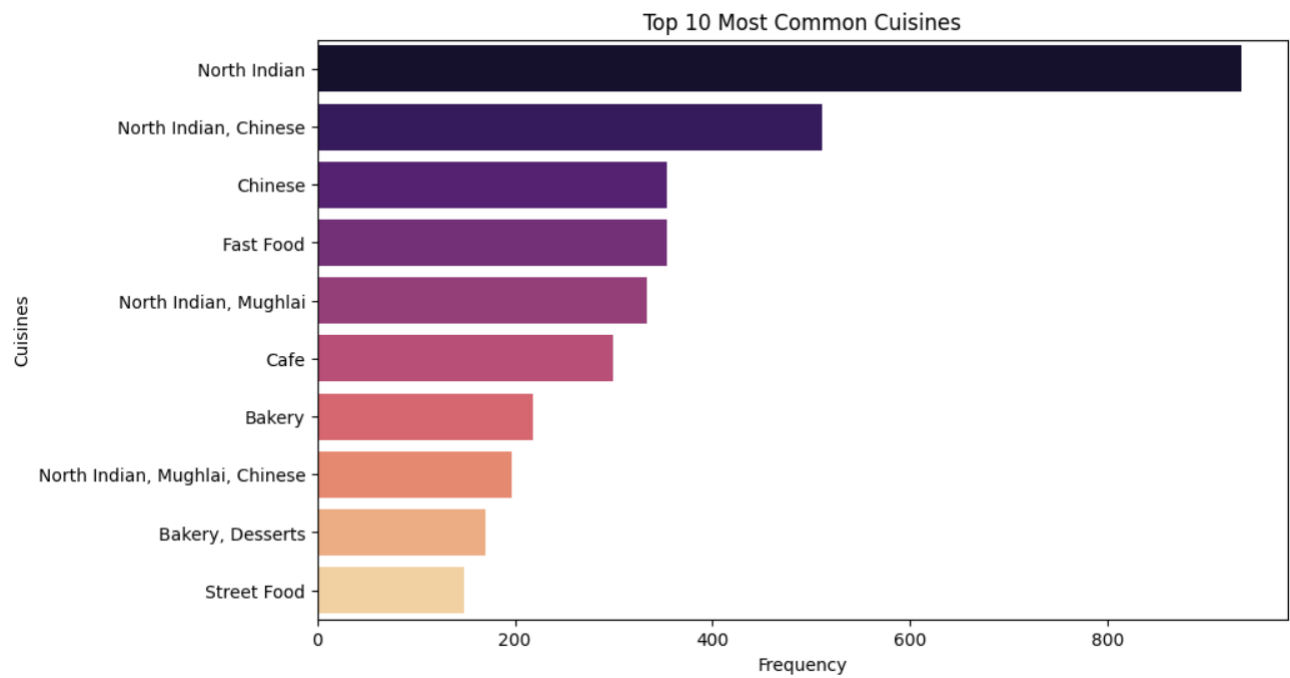


Fig6- Most Common Cuisines

Online Delivery Availability

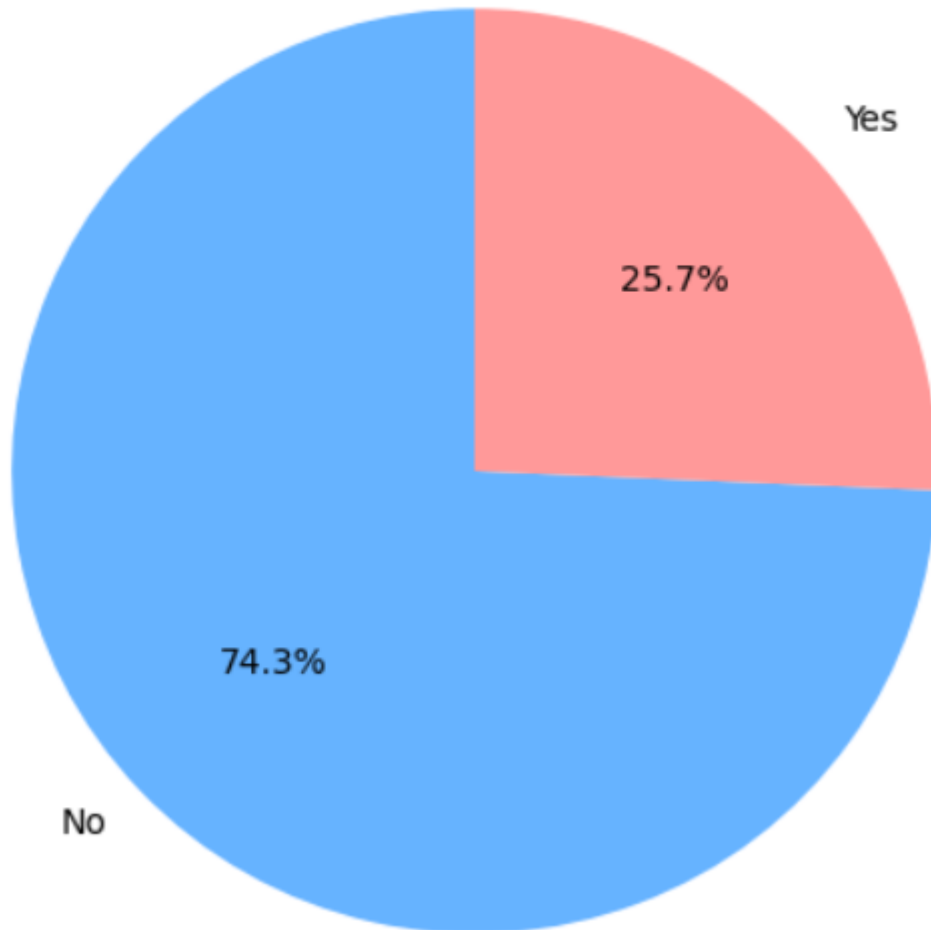


Fig7-Online Delivery Available

4.3 RESULTS AND OBSERVATION

- India is home to the majority of restaurants, especially in major cities like Mumbai, Bangalore, and New Delhi.
- Fast food, Chinese, and North Indian are the most popular cuisines.
- In India and the United Arab Emirates, online delivery services are fairly common.
- Average ratings are higher for restaurants that allow online ordering or reservations.
- Although they receive fewer votes, restaurants with greater price ranges typically have better reviews.
- According to the 'Average Cost for Two' feature, the most expensive restaurants were found in the UK, Australia, and Qatar.

Other Notes:

- Multi-cuisine restaurants enjoy more patronage.
- Fast food restaurants typically receive lower ratings but more votes.
- The distribution of restaurant types is dominated by cafes and casual dining.

5 CONCLUSION

This project explored the Zomato dataset using SQL and Python to understand the dynamics of the restaurant industry. Through structured querying and visualization, we discovered key insights on pricing, user preferences, and restaurant performance.

The analysis demonstrates how data can be leveraged to:

- Guide new restaurant openings
- Improve existing service offerings
- Enhance user experience on platforms like Zomato

In future work, incorporating sentiment analysis of reviews or time-series analysis of ratings could add depth to this study.

6 REFERENCES

- 1 Kaggle Zomato Dataset
- 2 Country Code Excel from Zomato
- 3 Python Documentation: pandas, seaborn, matplotlib
- 4 MySQL Reference Manual
- 5 Stack Overflow for troubleshooting and solutions